

Worksheet 1

Exercise 1 Suppose that the weather from one day to the next is described by a Markov chain on the 0, 1, 2 (0 for sunny, 1 for cloudy, 2 for rainy) whose transition matrix is given by

$$\begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/4 & 1/2 & 1/4 \\ 0 & 1/2 & 1/2 \end{pmatrix}.$$

It's Tuesday and cloudy. Determine

1. *The probability that the next three days will be sunny.*
2. *The probability that next Friday will be sunny.*

Exercise 2 A Markov chain on three states, say 0, 1 and 2, has as transition matrix

$$P = \begin{pmatrix} 1/4 & 3/4 & 0 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \end{pmatrix}.$$

Determine

1. *State classes and their types;*
2. *The n -step transition matrix; you may use the eigenvalues and eigenvectors of P .*
3. *The limit of this matrix when n goes to infinity.*

Exercise 3 Consider a four-state Markov chain $S = \{1, 2, 3, 4\}$ whose transition matrix is

$$P = \begin{pmatrix} 0 & 1/2 & 1/2 & 0 \\ 1/2 & 0 & 0 & 1/2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

1. *Draw the transition graph associated with this Markov chain.*
2. *Show that states 3 and 4 are absorbing. Are the others transient or recurrent?*
3. *Calculate the probabilities of the following trajectories as a function of the initial probabilities of each state (p_0, q_0, r_0, s_0) :*
 $(X_0 = 1, \forall n \geq 1 \quad X_n = 3), (X_0 = 1, X_1 = 2, \forall n \geq 2 \quad X_n = 4),$
 $(X_n = 1 \text{ if } n \text{ is even and } X_n = 2 \text{ if } n \text{ is odd}).$
Show that the path $(X_n = 1 \text{ if } n \text{ is even and } X_n = 2 \text{ if } n \text{ is odd})$ has probability zero.

4. Assume that the distribution between the four states is uniform at initial time $t = 0$. Calculate the distribution at time $t = 1$ and then at time $t = 2$. Same question if we start from an initial distribution $\pi_0 = (\frac{1}{2}, \frac{1}{2}, 0, 0)$.
5. Show that any initial distribution of the form $\pi_0 = (0, 0, r_0, s_0)$ with $r_0 + s_0 = 1$ is a stationary distribution. Are there any others?

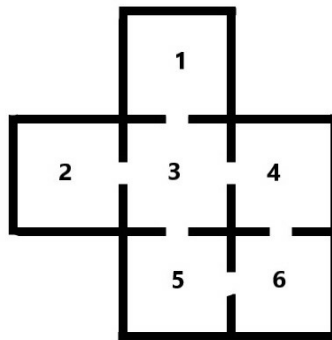
Exercise 4 The successive results of a player's chess games against chess software follow a Markov chain on the states W for win, L for loss and D for draw with the corresponding transition matrix given by

$$P = \begin{pmatrix} 3/4 & 0 & 1/4 \\ 0 & 3/4 & 1/4 \\ 1/2 & 1/4 & 1/4 \end{pmatrix}.$$

Determine:

1. The long-term proportion of wins for this player
2. The expected number of games from one win to the next.

Exercise 5 A rat runs through the maze shown below. At each step it leaves the room it is in by choosing at random one of the doors out of the room.



1. Give the transition matrix P for this Markov chain.
2. Show that it is irreducible but not aperiodic.
3. Find the stationary distribution
4. Now suppose that a piece of mature cheddar is placed on a deadly trap in Room 5. The mouse starts in Room 1. Find the expected number of steps before reaching Room 5 for the first time, starting in Room 1.
5. Find the expected time to return to room 1.

Exercise 6 A slot machine in a casino gives you 30 DZD if you win a game, but you have to pay 20 DZD per game to play. The machine advertises that there is always at least a 50/50 chance of winning each game. In fact, the probability of success in a game is equal to $\frac{k+1}{k+2}$, where k represents the number of successes in the two previous games. Is this slot machine profitable for the casino in the long term?